



μ Lab worksheet · one image, two jobs

Group _____ Date _____ Time _____

Driver (not the one who drove round 1) _____ Mapper _____

Hand this to the teacher when the round ends.

1 · THE ROUND, IN FOUR STEPS

1 ☐ Measure what you were given

`docker image ls oesteban/neuropipeline-302` — write the number down.

2 ☐ Build the notebook half

Jupyter and the viewer, plus `docker-cli`, and **none** of the neuroimaging stack. Measure that one too.

3 ☐ Give it the socket and a volume

`docker volume create project` then `run` the notebook with both `-v project:...` and `-v /var/run/docker.sock:/var/run/docker.sock`

4 ☐ Drive the tools from a cell

The notebook holds no tools. It starts a container that does.

2 · WHAT YOU WERE GIVEN

The image carries its own recipe. Read it without unpacking anything:

```
docker run --rm oesteban/neuropipeline-302 cat /opt/302/Dockerfile
```

Size _____ Layers(`image history`) _____

Name two things in that `Dockerfile` that a person **running** the notebook never needs, and say who does need them.

3 · THE HALF YOU BUILD

A Jupyter image with `docker-cli` in it and no neuroimaging software at all.

Its size _____ Times smaller _____

You added a tool to this image (`docker-cli`) in order to **remove** tools from it. Say in one sentence why that is not a contradiction.

4 • TWO KINDS OF COPY

`COPY` has two sources. Plain, it takes files from the **build context**: the directory you handed to `docker build`, on your own filesystem. With `--from`, it takes them out of an **image** instead, either one built earlier in the same `Dockerfile` (a **stage**) or one that already exists, like `docker:29-cli` here. Building an image out of pieces of other images is what makes a build **multi-stage**, and only the pieces you name cross over.

```
notebook/Dockerfile
FROM python:3.12-slim-trixie

# Copy from another image (??) the docker command line client
COPY --from=docker:29-cli /usr/local/bin/docker /usr/local/bin/docker

RUN pip install --no-cache-dir ipyniivue jupyter notebook

WORKDIR /work
COPY brain_mri_pipeline.ipynb .

EXPOSE 8888
CMD ["jupyter", "notebook", "--ip=0.0.0.0", "--port=8888", "--no-browser"]
```

Two `COPY` lines, one of each kind. For each, say where that file sat before the build started, and whether that place was on your laptop.

Now measure what the `--from` line cost you:

```
docker image pull docker:29-cli && docker image ls docker
docker image history notebook ← find the COPY --from line
```

The whole `docker:29-cli` image _____ The layer it gave you _____

Those two numbers are nowhere near each other. Say what crossed over and what stayed behind, and why `docker:29-cli` never runs on your machine.

Break it on purpose: comment that `COPY --from` line out, rebuild, and run `!docker ps` in a cell.

What comes back? _____ Then put the line back.

5 • GET IT WRONG FIRST, ON PURPOSE

From inside the notebook, make a directory and put a file in it. Then start a container mounting **that path** and look at what arrived:

```
mkdir -p /tmp/mine && echo hello > /tmp/mine/file
docker run --rm -v /tmp/mine:/mnt alpine ls -A /mnt
```

What did it list? _____ Was there an error? _____

The path exists. You just made it. So who went looking for it, and where did **they** look?

Now do it again with a **named volume** instead of a path, and say why that one works.

6 • DRIVING THE TOOLS FROM A CELL

Put the data in the volume first, then one container per step. Every line below starts a container, and every container is gone before the next begins.

```
W=/home/databot/work
T=oesteban/neuropipeline-302

# seed the volume, and its ownership, from the image itself
!docker run --rm -v project:$W $T sh -c "cp \${HOME}/data/ds000005/sub-01/anat/sub-01_T1w.nii.gz $W/"

!docker run --rm -v project:$W $T mri_synthstrip -i $W/sub-01_T1w.nii.gz -o $W/brain.nii.gz
!docker run --rm -v project:$W $T niimath $W/brain.nii.gz -unifize $W/uni.nii.gz
!docker run --rm -v project:$W $T tissue_segment $W/uni.nii.gz $W/dseg.nii.gz
!docker run --rm -v project:$W $T tissue_volumes $W/dseg.nii.gz $W/volumes.tsv
```

Write the three tissue volumes it reports. CSF _____ GM _____ WM _____

There is no neuroimaging software in your notebook image. So how did a brain get segmented? Answer in terms of **who ran what, where**.

7 • WHAT THE SPLIT COST, AND WHAT IT BOUGHT

Run `docker ps` in another terminal while a step is running, and again two seconds later. What is different, and what does that tell you about where the work happens?

Both containers mounted `project`. Neither could see the other's filesystem. So where does the brain image actually live between two steps?

Samira's replacement wants to change the segmentation tool next month. With the monolith, what would have to be rebuilt? With the split?

8 • THE THING YOU HANDED OVER

You gave a notebook `/var/run/docker.sock`. A notebook that runs whatever anyone types into it.

Name one thing a cell could now do that has nothing to do with neuroimaging.

This is fine on your own laptop this afternoon. Say the condition that would make it not fine.

Done when the notebook renders a brain it did not compute, and you can state both image sizes.